Exp. (2): Simple Pendulum Experiment

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1. Objectives

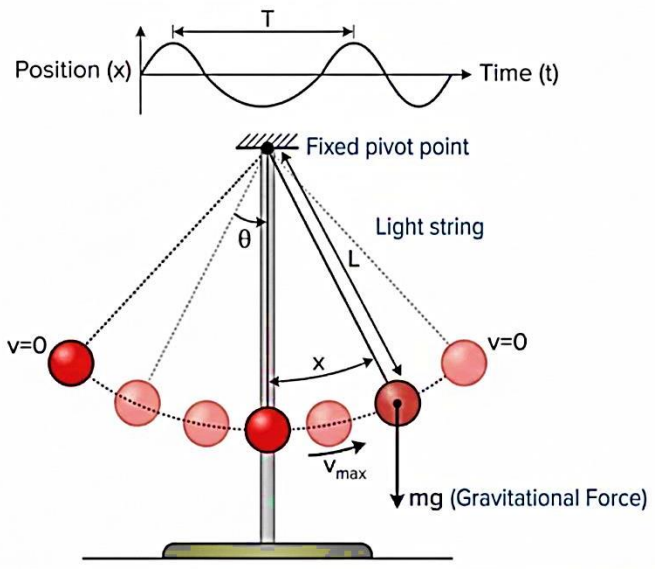
- Determining the acceleration due to gravity (g) experimentally using a simple pendulum setup.
- Investigating the relationship between pendulum length (L) and time period (T)

2. Apparatus

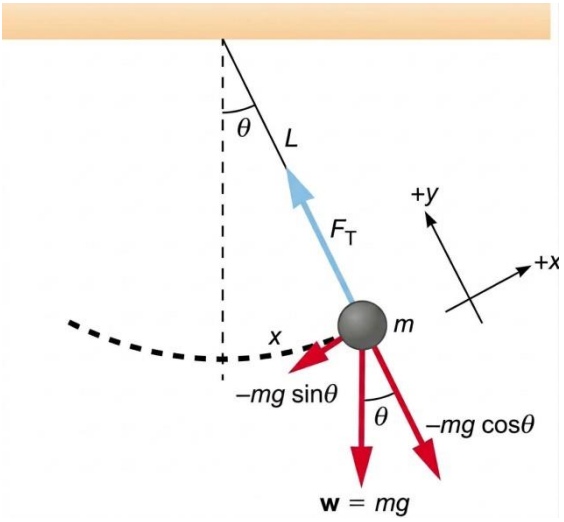
- Retort stand and clamp
- Light string
- Pendulum bob
- Meter ruler
- Stopwatch

3. Theory

A simple pendulum consists of a small bob suspended from a fixed point by a light, inextensible string. When the bob is displaced through a small angle and released without pushing, it oscillates about its equilibrium position.



When a pendulum is displaced by an angle θ from its vertical equilibrium position, the net force acting along the arc of motion is provided by the tangential component of gravity:



$$F = -mg \sin\theta \quad (1)$$

For large angles, this equation is non-linear and complex to solve. However, when restricted to **small angles** ($\theta < 5^o$), we can use the small-angle approximation where ($\sin\theta \simeq \theta$). Since the arc length is $x = L\theta$, the restoring force becomes directly proportional to displacement:

$$F = -mg \frac{x}{L} \quad (2)$$

$$F = -\left(\frac{mg}{L}\right) x \quad (3)$$

This linear restoring force is the exact condition required for **Simple Harmonic Motion**. The **Simple Harmonic Motion (SHM)** is a specific type of periodic motion where an object oscillates back and forth around an equilibrium position, governed by two strict conditions:

1. **A restoring force** acts on the object, always pulling or pushing it back toward the central equilibrium position.
2. **The magnitude of this restoring force** is directly proportional to the object's displacement from equilibrium.

Simple Harmonic Motion (SHM) is fully defined and characterized by a specific set of **kinematic parameters.** These parameters describe the geometry and timing of the oscillation:

 Amplitude (A): The maximum displacement of the object from its equilibrium position.

 Period (T): The time required to complete one full oscillation cycle (back and forth).

 Frequency (f): The number of complete cycles completed per unit of time.

Solving the differential equation (3) for this system yields the theoretical formula for the period of oscillation (T), which is the time required to complete one full back-and-forth cycle:

$$T = 2\pi\sqrt{\frac{L}{g}} \quad (4)$$

where:

- T: is the period (Periodic Time) (s)
- L: is the pendulum length, measured from the suspension point to the center of the bob (m)
- g: is the acceleration due to gravity (m/s^2)
- f: is the frequency (Hz)

Squaring both sides gives:

$$T^2 = \frac{4\pi^2}{g} L \quad (5)$$

This equation has the form of a straight line, (y=mx). Therefore, plotting (T^2) against (L) produces a straight line whose slope is:

$$Slope = \frac{4\pi^2}{g} \quad (6)$$

Hence, the acceleration due to gravity can be determined from:

$$g = \frac{4\pi^2}{Slope} \quad (6)$$

4. Experimental Procedure

1. Set up the retort stand and suspend the pendulum bob using the string.
2. Measure the pendulum length (L) from the suspension point to the center of the bob.
3. Displace the bob through a small angle of approximately (5°)–(10°).
4. Release the bob gently without pushing it.
5. Measure the time (T_{10}) required for 10 complete oscillations using a stopwatch.
6. Calculate the period using (T=t/N), where (N=10).
7. Calculate (T^2) for the measured length.
8. Repeat the experiment for at least five different pendulum lengths.
9. Record the results and plot (T^2) against (L).

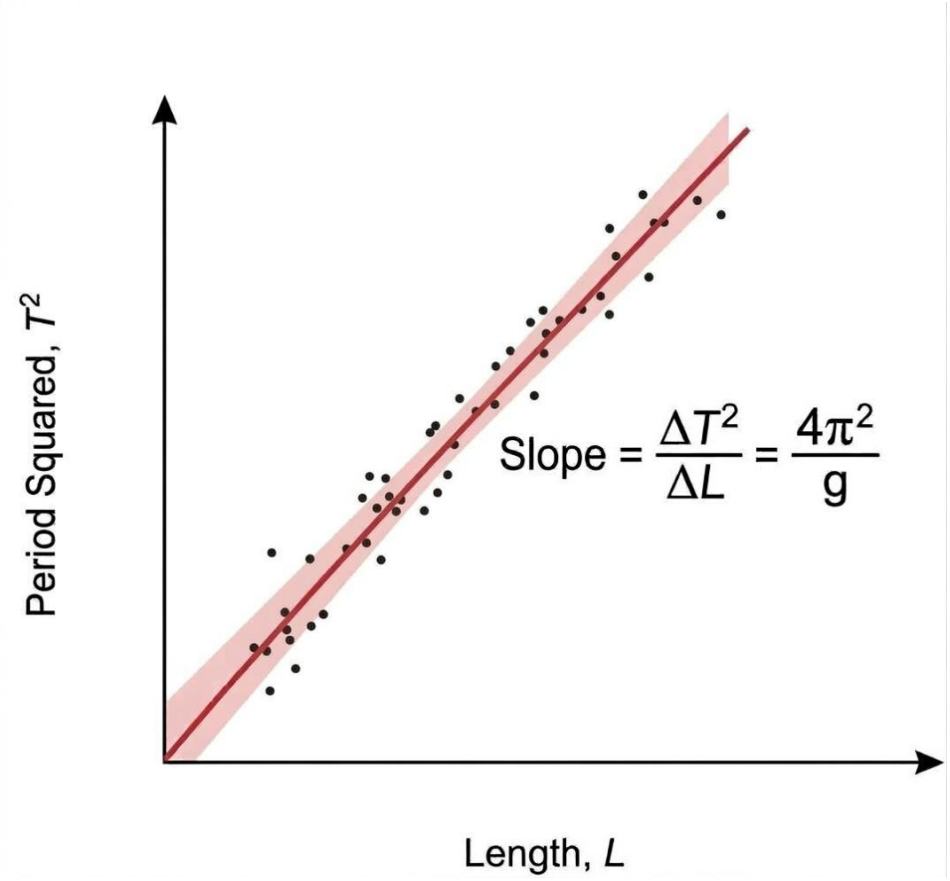
5. Data Collection

Record your observations in the following table:

No.	(L) (cm)	T_{10} (s)	($T=T_{10}/10$) (s)	(T^2) (s^2)
1				
2				
3				
4				
5				

6. Results and Discussion

Record the final result obtained from your graph.



- Experimental acceleration due to gravity:
  - $g_{exp.} = \frac{4\pi^2}{Slope}$
- Reference value:
  - $g_{ref.} = 9.81\ m/s^2$, accepted value of gravitational acceleration.

- **The percentage error in the experimental value of gravitational acceleration (g) is calculated using:**

$$Error = \frac{|g_{ref.} - g_{exp.}|}{g_{ref.}} \times 100$$

- **Discuss possible sources of error, including:**

- Human reaction time when using the stopwatch.
- Incorrect measurement of the pendulum length.
- Excessive initial angular displacement.

- Air resistance and friction at the pivot.
- Incorrect counting of complete oscillations.

7. Review Questions

1. Why should the pendulum be released from a small angle?
2. Why is the pendulum length measured to the center of the bob?
3. Why is it preferable to time 10 oscillations rather than one?
4. Why is a graph of (T^2) against (L) a straight line?
5. How can the slope of the graph be used to determine (g)?